



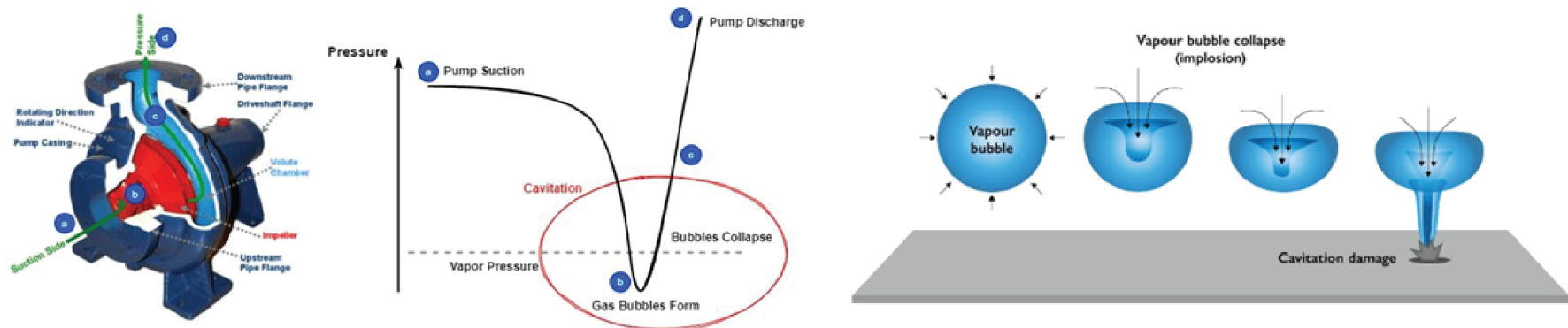
A NUMERICAL APPROACH TO STUDYING CAVITATION IN OIL AND GAS INDUSTRY

**OKENE DANIEL OMONIGUHNO
MADU CHUKWUEMEKA MIRACLE
CHUKWUEZE BENJAMIN NWABUEZE**

Supervised By:
Dr Owunna Ikechukwu



Introduction



Cavitation, a classic gas–liquid two-phase flow phenomenon which has been studied by various researchers and has been observed to be the leading factor for the failure of many rotating machineries just like the HMS Daring. In the context of the oil and gas industry, the significance of cavitation extends beyond equipment damage. Cavitation can disrupt the flow of crude oil, leading to interruptions in production and transportation (Li et al., 2023), which in turn impacts the profitability of operations.

As the global demand for energy grows and the reliance on oil remains strong, there is a need to optimise every aspect of crude oil transportation systems, particularly the pumping mechanisms. Pumps that fail due to cavitation can lead to costly repairs, downtime, and environmental hazards like spills. Therefore, understanding the dynamics of cavitation when crude oil is the working fluid, and finding solutions to mitigate its effects, is critical for improving pump reliability and operational efficiency.

Abstract

The oil and gas industry relies heavily on centrifugal pumps to transport crude oil, but cavitation remains a significant challenge, reducing both the lifespan and efficiency of pumps. Despite its impact on operations, cavitation in crude oil has received limited attention, as most existing studies have focused on water and other fluids. This study aims to bridge this knowledge gap by developing a numerical approach to analyzing cavitation in crude oil centrifugal pumps. We employed computational fluid dynamics (CFD) simulations using the ANSYS (R12025) tool to investigate the complex flow patterns and cavitation phenomena in centrifugal pumps. Our approach was multi-faceted to accurately capture the complexities of cavitation. We used a bubble model to track the nucleation, growth, and implosion of vapour bubbles, a turbulence model to simulate chaotic flow behavior, and a multiphase model to account for interactions between crude oil in its liquid and vapour phases. The numerical approach was validated by comparing our results with actual performance data from a pump used on an established oil rig, yielding a margin of error of approximately 0.5%. The results underscore the importance of considering crude oil properties and pump design parameters in mitigating cavitation. Notably, the roughness value of the pump casing was found to have a significant impact on cavitation, with even slight changes leading to increased cavitation activity. Furthermore, crude oil backflow was identified as a key factor contributing to discharge cavitation, highlighting the need for careful pump design and operational strategies in the oil and gas industry. mitigation strategies.



Aim and Objectives

The primary aim of this study is to employ Computational Fluid Dynamics (CFD) to examine cavitation effects on centrifugal pump performance and flow patterns in oil and gas applications using ansys simulation tool.

The specific objectives of the study are:

- To simulate cavitation in a centrifugal pump handling crude oil using CFD techniques, focusing on identifying regions where cavitation is likely to occur.
- To analyse the influence of crude oil properties (viscosity, density, and impurities) on cavitation dynamics, examining how these properties affect the formation, growth, and collapse of cavitation bubbles



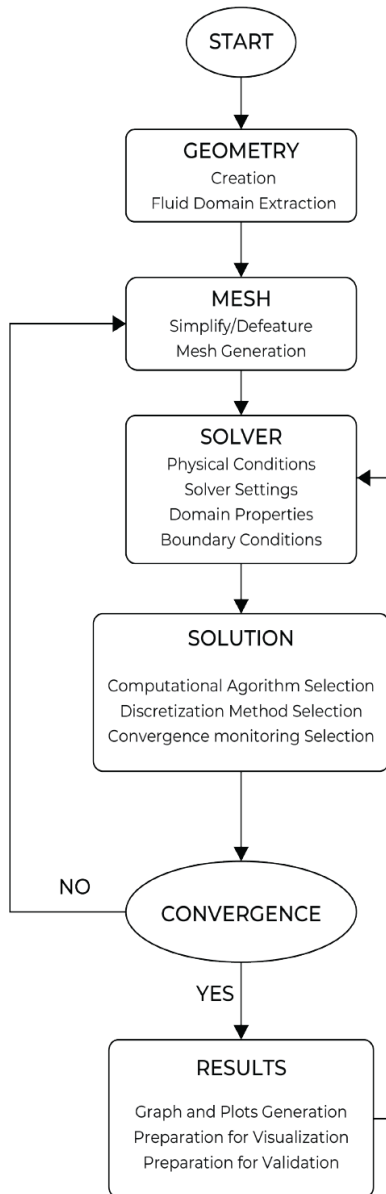
Literature Review



Although cavitation was first investigated by Barnaby and Parsons in 1893, which arose during the analysis of performance problems in British warships, HMS Daring, its conception and the effect has been studied over the years by various researchers in various field. Some of which includes:

- i) Cui and Li: In 2010 investigated internal flow characteristic of energy loss in high-speed centrifugal pumps by employing the SST $k-\epsilon$ turbulence model and the results obtained at the course of their investigations revealed that the energy loss in the impeller reduced as the axial distance between the inducer and impeller increased. Their study was based on understanding the effect of turbulence of the fluid on cavitation conception.
- ii) Hu et al: In 2011 analysing the occurrence of cavitation in an overloaded condition using the ZBG model reported that at overloaded conditions, flow separation, characterised by unsteady vortex shedding from the tongue, causes the low pressure zone near the tongue.
- iii) Ali Abdul: in 2014 computationally investigated the effect of cavitation on centrifugal pumps outlines that the percentage efficiency drop of the pump is greater than 3% and that the cavitation regions appear at the leading edge of suction side of the impeller blades which represents the lowest pressure area inside the computational domain of the centrifugal pump
- iv) Corvaglia et al: In 2016, while studying the effect of cavitation, evaluated the impact of tooth space pressure in case of incomplete filling, reported a fall in the flow rate when a lower pressure values were reached. This study explored the relationship between the geometry of the impeller (particularly the tooth space) and how it affects the onset and severity of cavitation, as well as the overall efficiency and reliability of the pump.

Methodology



Theories of Fluid Flow

I. Conservation of Mass: The Continuity Equation

The Continuity Equation ensures that mass is conserved in a fluid flow system. For a control volume, this principle is expressed as:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{U}) = 0 \dots \dots \dots (3.0.1.1)$$

III. Conservation of Energy: The Energy Equation

The Energy Equation accounts for energy transformations within a fluid system. In differential form, for incompressible flows, it is expressed as:

$$\frac{\partial}{\partial t} \left(\frac{\rho \vec{U}^2}{2} + \rho e + \rho \Phi \right) + \nabla \cdot \left(\vec{U} \left(\frac{\rho v \vec{U}^2}{2} + \rho e + \rho \Phi + p \right) - \kappa \nabla T \right) = 0 \dots \dots \dots (3.0.1.4)$$

II. Conservation of Momentum: The Navier-Stokes Equations

The Navier-Stokes equations describe the motion of fluid particles by applying Newton's second law to a fluid element. For an incompressible, Newtonian fluid like crude oil, the equations take the form:

$$\rho \left(\frac{\partial \vec{U}}{\partial t} + \vec{U} \cdot \nabla \vec{U} \right) = -\nabla p + \mu \nabla^2 \vec{U} + f \dots \dots \dots (3.0.1.3)$$

VOF Model

The volume-fraction of a phase is given by:

$$C = \frac{\text{Volume of cell grid occupied by the phase}}{\text{Total volume of cell grid}} \dots \dots \dots (3.0.2.1)$$

The volume-fraction with respect to the characteristic function is given as:

$$C = \frac{1}{\Delta V} \oint H(x, y, z, t) dV \dots \dots \dots (3.0.2.2)$$

Where: $H(x, y, z, t) \rightarrow$ Characteristic function, $C \rightarrow$ Volume - fraction, $\Delta V \rightarrow$ Volume of cell grid

In order to accurately account for the motion of the fluid and change in interface at a given interval, VOF model uses the following transport equations:

I. Momentum Transport Equation

$$\frac{\partial (\rho \vec{U})}{\partial t} + \nabla \cdot (\rho \vec{U}) = -\nabla P + \nabla \cdot (\mu (\nabla \vec{U} + \nabla \vec{U}^T)) - F_s \dots \dots \dots (3.0.2.4)$$

Where: $\rho \rightarrow$ Density, $P \rightarrow$ Pressure, $\mu \rightarrow$ Coefficient of viscosity,

$\vec{U} \rightarrow$ Velocity profile, $F_s \rightarrow$ Tension force,

II. Volume Fraction Transport Equation

$$\frac{\partial C}{\partial t} + \nabla \cdot (C \vec{U}) + \nabla \cdot (C(1-C)\mu) = 0 \dots \dots \dots (3.0.2.13)$$

Bubble Model

3.0.3 Cavitation Bubble Dynamics: Schnerr-Sauer Model

Cavitation simulation involves the formation, growing and the collapsing of bubbles. In order to track the growth and collapsing of bubbles in the simulation, computational fluid dynamic (CFD) solvers (Ansys fluent in our case) use various bubble transport equation e.g. Schnerr-Sauer, Zwart-Gerber-Belamri etc. depending on the set up. This bubble equations are derived from the generalized Rayleigh-Plesset equation, given as:

$$\frac{P_{vap}(t) - P_{\infty}(t)}{\rho_l} = R_b \frac{d^2 R_b}{dt^2} + \frac{3}{2} \left(\frac{dR_b}{dt} \right)^2 + \frac{4\gamma_1}{R} \frac{dR_b}{dt} + \frac{2F}{\rho_l R_b} \dots \dots \dots (3.0.3.1)$$

For the simulation of the phenomenon of cavitation formed during the transport of crude oil and in order to accurately track the formation, growth, collapse, and the volume occupied by the bubble, Schnerr-Sauer cavitation model is used as the bubble transport equation.

The Schnerr-Sauer cavitation model is given by:

$$R \frac{d^2 R}{dt^2} + \frac{3}{2} \left(\frac{dR}{dt} \right)^2 = \frac{P(R) - P_{\infty}}{\rho_l} - \frac{2F}{\rho_l R} - \frac{4\mu}{\rho_l R} \frac{dR}{dt} \dots \dots \dots (3.0.3.2)$$

The growth of bubble radius is defined as:

$$\frac{dR}{dt} = \rho_{vap} \frac{\rho_l}{\rho} C(1-C) \frac{3}{R_b} \sqrt{\frac{2(P_{vap} - P)}{\rho_l}} \dots \dots \dots (3.0.3.4)$$

Where:

$P \rightarrow$ Local fluid pressure

The bubble radius (R_b) for Schnerr-Sauer cavitation model is given by:

$$R_b = \left(\frac{C}{1-C} \frac{3}{4\pi n} \right)^{\frac{1}{3}} \dots \dots \dots (3.0.3.5)$$

The volume of fraction used in Schnerr-Sauer cavitation model is given by:

$$C = \frac{n_0}{1 + n_0 \frac{4}{3} \pi R^3 \frac{d}{dt} \left(\frac{4}{3} \pi R^3 \right)} \dots \dots \dots (3.0.3.6)$$

Turbulence Model - RNG k-ε Model

- ii. The **dissipation rate (ε)** represents the rate at which turbulent kinetic energy is converted into thermal energy due to viscous dissipation, and its equation is given by:

$$\frac{\partial \varepsilon}{\partial t} + U_j \frac{\partial \varepsilon}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\left(\nu + \frac{\nu_\tau}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x_j} \right] + C_{\varepsilon 1} \frac{\varepsilon}{k} \tau_{ij} \frac{\partial U_i}{\partial x_j} - C_{\varepsilon 2} \frac{\varepsilon^2}{k} \dots \dots \dots (3.0.4.2)$$

The k-ε model is a two-equation turbulence model that solves two transport equations: one for the turbulent kinetic energy (k) and another for its dissipation rate (ε). This approach allows the model to account for the transport of turbulence properties and their interaction with the mean flow.

- i. The **turbulent kinetic energy (k)** represents the intensity of turbulence, and its equation is given by:

$$\frac{\partial k}{\partial t} + U_j \frac{\partial k}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\left(\nu + \frac{\nu_\tau}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + \tau_{ij} \frac{\partial U_i}{\partial x_j} - \varepsilon \dots \dots \dots (3.0.4.1)$$

I. RNG k-ε Model

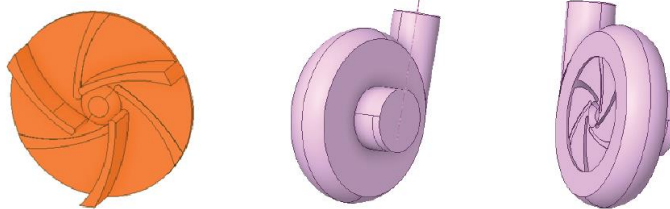
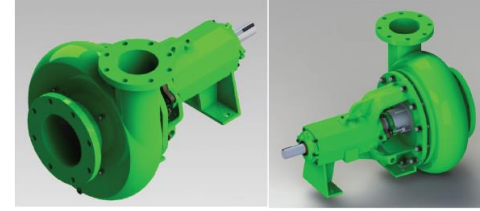
The RNG k-ε model (Re-Normalization Group), a more recent version of the k-ε model developed by Yakhot and Orszag (1986) which enhances the standard k-ε model by incorporating additional terms derived from renormalization group theory, improving accuracy in rapidly strained and swirling flows. Similar to the standard model, the eddy viscosity, k and ε are still given by Equations (3.0.4.1) and (3.0.4.2). However, the model uses a modified coefficient, $C_{\varepsilon 2}$, defined by

$$C_{\varepsilon 2} \equiv \bar{C}_{\varepsilon 2} + \frac{C_\mu \lambda^3 \left(1 - \frac{\lambda}{\lambda_0} \right)}{1 + \beta \lambda^3}, \quad \text{where,} \quad \lambda = \frac{k}{\varepsilon} \sqrt{2 S_{ji} S_{ij}}$$

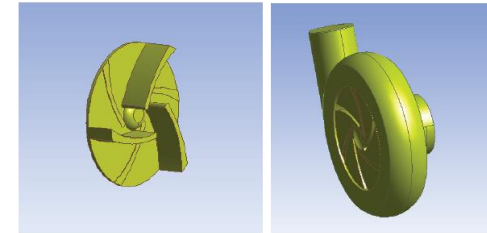
Methodology



Geometry Creation



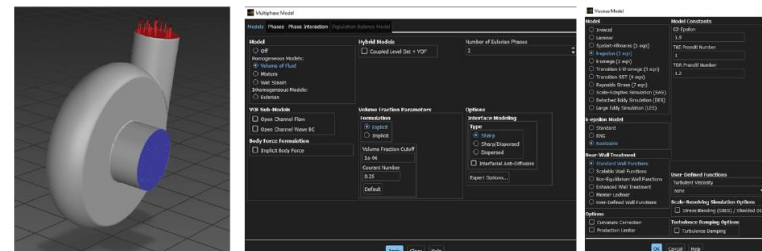
Fluid Domain Extraction



Defeaturing



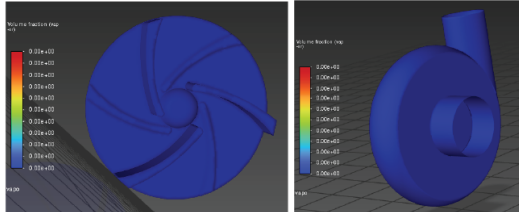
Mesh Generation



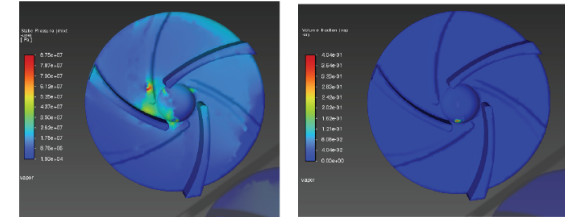
Solver Setup and Boundary Conditions

Methodology

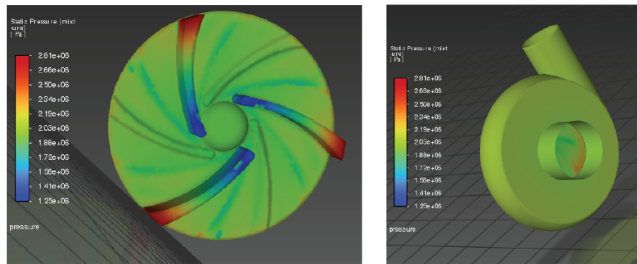
Liquid and Vapour



Liquid, Air and Vapour

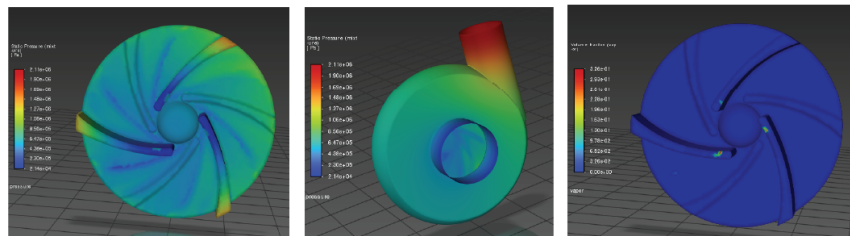


Iteration I: No roughness Value



Iteration II: Roughness Value of 0.00015(m)

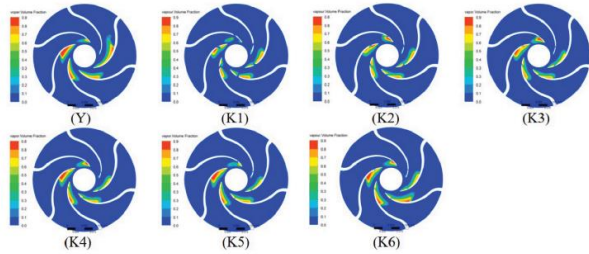
Running a second iteration with roughness value of 0.00015(m) and backflow VOF of air as 0.2 and Crude Vapour as 0.8. It can be seen that there is little cavitation at the impeller eye and pressure drop compared to the first iteration.



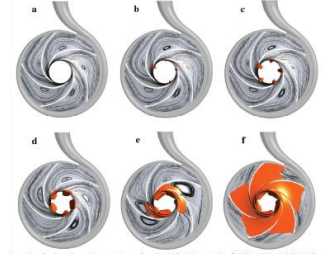
Iteration III: Roughness Value of 0.0015(m)

Running a third iteration with roughness value of 0.0015(m) and backflow VOF of air as 1. It can be seen that there is more cavitation at the impeller eye and pressure drop compared to the second iteration.

Result, Discussion & Conclusion



F. Orlandi et al (2023)



Wu et al. Chin. J. (2019)

Literature Validation

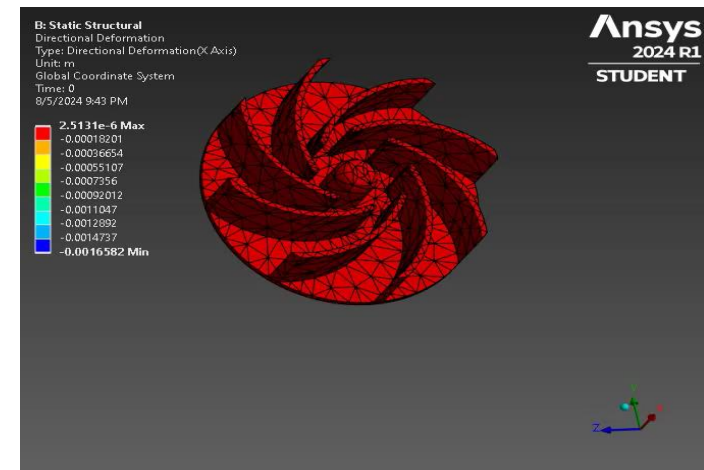
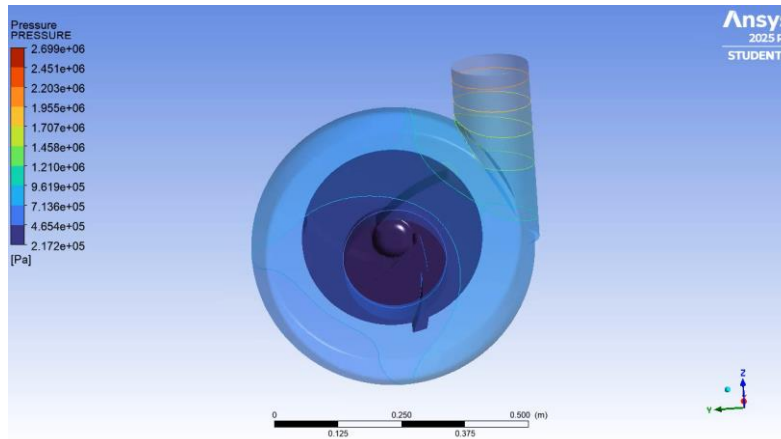
Looking at existing literature validates our analysis as cavitation occur in similar parts of the impeller and casing.



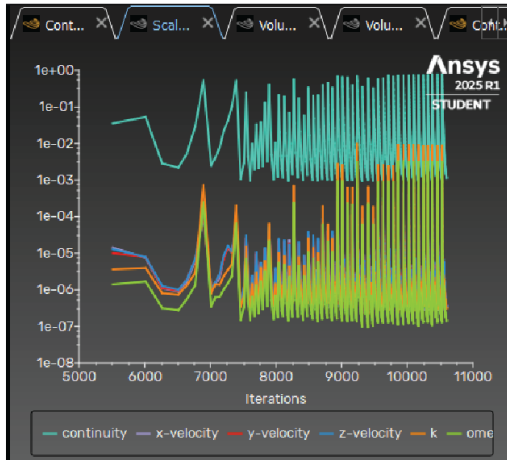
Experimental Validation

Validating our results with images provided by the data provider for the simulation validates our results as well

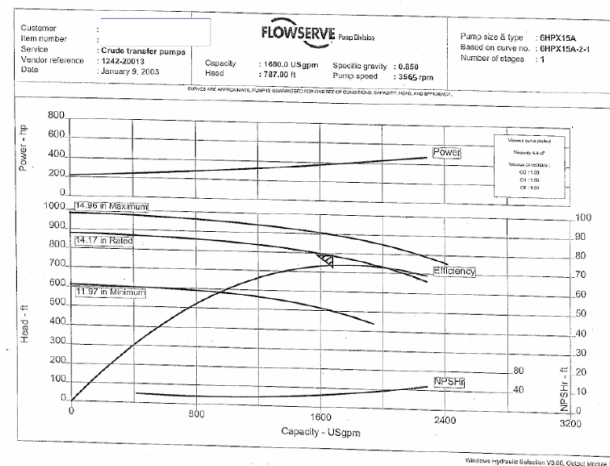
Effects



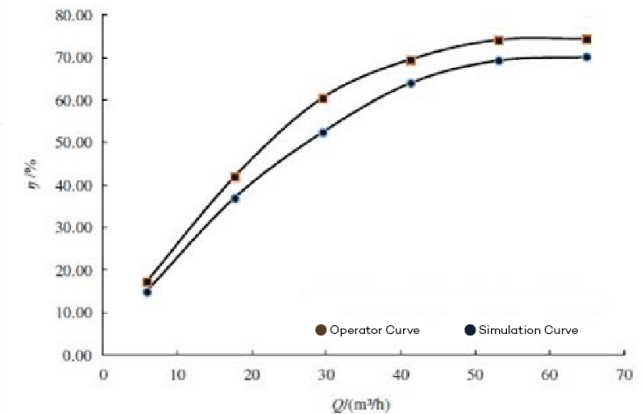
Result, Discussion & Conclusion



Convergence Graph



Pump Performance Comparison



Conclusion

From the transient analysis carried out, it is concluded that crude-oil cavitation inception increases drastically due to increase in the backflow of air [causing an increase in the vortex formation due to air-lock occurring in the pump) and increase in roughness value of the casing. Also it is seen that cavitation effects goes beyond damage to the impeller alone, but also to bearings and seals of the pump solely due to vibration.

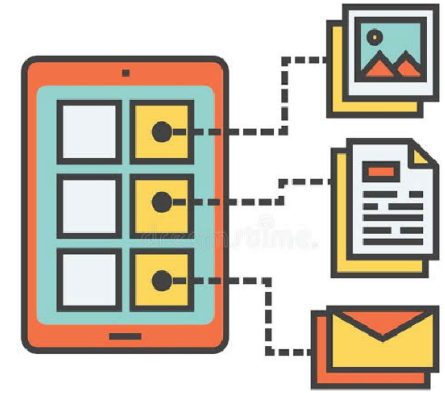
Recommendation

Based on the analysis conducted, the following are recommendation to crude-oil pump operators

- Factor cavitation when ordering or designing pumps for use.
- The use of booster pumps before mainline pump so as to keep pressure higher than vapour pressure
- Operating with parameters that prioritize efficient performance, yet with less cavitation effect.

References

- Li, C., He, J., Jia, W., Yang, F., Ban, J., & Qiu, B. (2023). Modeling transient cavitating flow in large drop crude oil pipelines. *Journal of Petroleum Science and Engineering*, 220. <https://doi.org/10.1016/j.petrol.2022.111241>
- Orlandi, F., Montorsi, L., & Milani, M. (2023). Cavitation analysis through CFD in industrial pumps: A review. In *International Journal of Thermofluids* (Vol. 20). Elsevier B.V. <https://doi.org/10.1016/j.ijft.2023.100506>
- Mousmoulis, G., Karlsen-Davies, N., Aggidis, G., Anagnostopoulos, J., & Papantonis, D. (2017). Experimental analysis of the onset and development of cavitation in a centrifugal pump. *Journal of Physics: Conference Series*, 813(1). <https://doi.org/10.1088/1742-6596/813/1/O12044>
- Luo, H., Wen, J., Lv, C., & Wang, Z. (2022). Modeling of viscosity of unstable crude oil–water mixture by characterization of energy consumption and crude oil physical properties. *Journal of Petroleum Science and Engineering*, 212. <https://doi.org/10.1016/j.petrol.2022.110222>
- Ravikumar, V. (2023). Pumps. In *Sprinkler and Drip Irrigation* (pp. 597–639). Springer Nature Singapore. https://doi.org/10.1007/978-981-19-2775-1_23





**THANK
YOU!!!!**